This document will give a simple overview of the data analysis methods that have been implemented and that will be built upon in the future. All of the current analysis scripts can be found in the Clinical Trial Scripts folder.

Step-Counting Algorithm

This script is titled "Pedometer_Script_Classification_Trial". Its purpose is to be given a continuous section of accelerometer data and count the number of steps a patient has taken. It takes a filepath and timestamps (in seconds since midnight) to mark the beginning and end of the relevant data. This data is then converted into SI units and the magnitude of the signal is calculated. This magnitude is then filtered for relevant walking speeds. A peak-picking algorithm is used to determine the number of steps taken, with each peak counting as two steps (the device is only worn on one foot). Height, prominence, and distance thresholds can be set to determine which peaks are counted or skipped. From there, a gait confirmation analysis is used to make sure that only peaks that are caused by walking are counted, while discarding isolated peaks. The final number of peaks is multiplied by 2 and reported as the step count. A visual representation of the data is also generated.

Feature Extraction Algorithm

This script is titled "Trial_Classification_Test". Its purpose is to take raw data from a wrist and ankle file, calculate relevant features for each time point to later be used in classification models, and create a resultant dataset. A fuzzy join is used to time-sync the provided wrist and ankle files. These files are then analyzed in 5-second sliding windows with a 50% overlap. For each window, the following 20 features are extracted:

- Wrist and Ankle (duplicated):
    - Signal Magnitude Area (SMA)
    - Acceleration Magnitude Mean
    - Acceleration Magnitude Standard Deviation
    - Acceleration Magnitude Root Mean Squared Error
    - Acceleration Magnitude Spectral Energy
    - Acceleration Magnitude Spectral Entropy
    - Angular Acceleration Standard Deviation
    - Jerk Magnitude Standard Deviation

        - Angular Jerk Magnitude Standard Deviation
- Combined:
    - Acceleration Magnitude Cross-correlation
    - Angular Acceleration Magnitude Cross-correlation

The resulting dataset has rows with all 20 calculated features and the timestamp (in seconds since midnight) of the window that they are associated with.

<u>Classification Algorithm</u>

This script is titled “Activity_Classification_Models”. Its purpose is to take the dataset of extracted features after it has been manually classified (see Data Processing Notes section) and create/test ML models to classify the data. The two models that are used in this script are a random forest classifier and a gradient boosting classifier. The file_pattern variable can be modified to filter for specific datasets in the relevant directory. Data from all of the relevant files are concatenated, and the extracted features are used for the x variables while the exercise type is used for the y variable. Data is then split and scaled and the models are created/tested. Confusion Matrices from the best version of each model are displayed to the user once the model is done running, along with the test accuracy of this model. A graph is also generated comparing the magnitude of a few features during different exercises. This was used to verify that the data that was being used made physical sense.

Data Processing Notes

Raw Data Processing - The raw ankle and wrist data should be slightly modified before running the feature extraction script. The feature extraction script uses timestamps in seconds since midnight, whereas the raw data reports milliseconds since midnight. An extra column (labeled “Time(s)” should be added that converts these timestamps to seconds. Additionally, it can be useful to add another column that converts this reading to real time (labeled “RWT”). This can help to verify that the data in the file is from the correct time, and to identify relevant jumps in time. This conversion can be done by taking the timestamp in seconds and dividing by 86400, then formatting the cell in standard time.

Manual Classification - After extracting features from the data, the activity windows were manually classified by the team. This process involved watching timestamped exercise videos from the patient, and labeling each window that a patient was performing an exercise in. The resulting dataset had each window marked with the 20 extracted features, along with a label of the exercise they were performing during that window (or no label if they were not performing an exercise at that time).